Orthogonally Invariant Ising Pressure: Certified Spectral Bins and Quantitative Constrained-Block Replacement

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Abstract

We study the stated all-temperature variational formula with its uniform spin probability, Haar conditional law, spectral edges and ordered field supremum retained. Two local refinements are proved. First, bin barycenters and endpoint mixtures bracket the exact edge-extended spectral transform and hence the stated variational values. Exact resolvent remainder identities give quadratic bin-width errors under a verifiable normalized gap, including a refinement by known within-bin variances and overlap-dependent factors. The order brackets themselves need no gap. Second, the magnetic proof's constrained-block replacement is sharpened: its depth-uniform Gaussian tail bound yields loss $O(\sqrt{n}\log n)$ instead of the displayed $O(n^{3/4}+\sqrt{n}\log n)$, and a direct entropy bound gives an explicit $O(n^{-1/4}\sqrt{\log n})$ two-spin error. The height cap and interior magnetizations are fixed. These results do not independently validate the source's Haar, overlap, cavity or all-temperature equality steps, and do not provide a rate in the original system size. Classical convexity, resolvent algebra and entropy inequalities are credited as such. Contribution assessment: C3/5, two source-specific quantitative refinements with certified spectral brackets and a sharper uniform constrained-block error, while the global pressure identification remains an external obligation.

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1 Model, conventions and proof dependencies

Let Λ_N be deterministic real diagonal, let U_N be Haar on $O(N)$, and put $J_N = U_N^{\mathsf{T}} \Lambda_N U_N$. The source [1] uses

$$P_N(c) = \frac{1}{N} \log \left[2^{-N} \sum_{\sigma \in \{-1,1\}^N} e^{\sigma^{\mathsf{T}} J_N \sigma/2 + c\cdot\sigma} \right]. \tag{1.1}$$

Temperature is absorbed into Λ_N and c. Counting spin measure adds $\log 2$ to this pressure. Removing the diagonal of J_N subtracts $\operatorname{Tr}\Lambda_N/(2N)$ exactly; it is not an alteration of the overlap law. Physical free energy at inverse temperature $\beta > 0$ has the opposite sign and a factor $1/\beta$.

The source assumes weak convergence of the empirical spectral law to a compact probability law μ, and convergence of the two extreme eigenvalues to the respective support edges. Weak convergence or finitely many moments alone is not that assumption. In the random-data extension the orbit law must be Haar *conditional jointly on the spectrum and external field*. Uniform integrability is separately needed for expectation limits.

Write $e_\mu = \max \operatorname{supp} \mu$. For $x > 0$ define $b_\mu(x) > e_\mu$ by

$$H_\mu(b_\mu(x)) = x, \qquad H_\mu(z) = \int \frac{\mu(\,d\lambda)}{z - \lambda}, \tag{1.2}$$

when such a solution exists, and otherwise set $b_\mu(x) = e_\mu$. Then $R_\mu(x) = b_\mu(x) - 1/x$ and $R_\mu(0) = \int \lambda \mu(\,d\lambda)$. The edge branch is part of the definition, not analytic continuation.

Let $\mathcal{P}$ be nondecreasing measurable $p : (0,1) \to [0,1]$ modulo null sets, and let $\mathcal{H}$ be nonnegative nondecreasing finite step paths. Put

$$D_p(r) = 1 - r - \int_0^1 (p(s) - r)_+ \, ds, \qquad 0 \le r \le 1. \tag{1.3}$$

For step h, take interval weights $w_i > 0$, cumulative masses $\zeta_i = \sum_{j<i} w_j$, levels h_i, and $h_{-1} = 0$. Starting with $F_{k-1}(z) = \log\cosh z$, set

$$\begin{aligned} F_{i-1} &= T_{\zeta_i, h_i - h_{i-1}} F_i, \qquad \ell(h) = F_{-1}(0) - h_{k-1}/2, \\ T_{a,v} f(z) &= \begin{cases} a^{-1} \log \mathbb{E} e^{a f(z + \sqrt{v} G)}, & a > 0, \\ \mathbb{E} f(z + \sqrt{v} G), & a = 0, \end{cases} \qquad G \sim N(0,1). \end{aligned} \tag{1.4}$$

In particular the root is an ordinary expectation. The proposed zero-field variational value is

$$\mathcal{F}_\mu = \inf_{p \in \mathcal{P}} \{ S(p) + G_{1,\mu}(p) \}, \quad S(p) = \sup_{h \in \mathcal{H}} \left\{ \ell(h) + \tfrac{1}{2} \int ph \right\}, \quad G_{t,\mu}(p) = \tfrac{1}{2} \int_0^1 t R_\mu(t D_p(r)) \, dr. \tag{1.5}$$

The order of infimum and supremum is not exchanged. The source already proves the covariance-level concavity of ℓ, its supporting inequality, L^1 Lipschitz constant $1/2$, and finiteness of $S(p)$ unless $p = 1$ a.e. Those results are not contributions of this report. In particular $S(0) = 0$.

Source file	Mechanism and retained scope
10-arrays	Joint GG identities, positivity, ultrametricity, synchronization, cascade marking, finite-replica continuity and Gaussian differentiation.
20-field	Tilted conditional magnetizations, weighted Hessian argument, concavity, support planes and trial domain.
30-rotation	Perturbations, Haar concentration and Ward equations; projected-overlap paths include their diagonal and root, not symbols alone.
40-upper	Fixed-trial contact comparison and tail order; preserves the field supremum inside the overlap infimum.
50-cavity	Base-law-preserving coupling, conditional fresh frames, noncommuting quadratic Gaussian tilt, cutoffs, site tests and self-consistency.
60-general	Qualitative edge-transform continuity, multiplicity comparison, spectral bins, random spectra, ground-state squeeze and counterexample.
70-magnetic	Magnetization conjugates, constrained blocks, depth-uniform mark laws, entropy comparison and field-law W_1 approximation.

All ten cached public texts were freshly read, including the complete main file, bibliography and README. This maps the manuscript; it is not an independent audit of all deep inputs or every global limiting argument.

2 Spectral bins that preserve order and first moments

Fix a compactly supported probability μ. Decompose it into finitely many subprobabilities μ_j of mass $\rho_j > 0$, supported in closed intervals $[l_j, u_j]$. Assign shared endpoints consistently so that $\mu = \sum_j \mu_j$. Let $m_j = \rho_j^{-1} \int \lambda \mu_j(\, d\lambda)$. For $l_j < u_j$, put

$$\mu^- = \sum_j \rho_j \delta_{m_j}, \qquad \mu^+ = \sum_j \rho_j \left[\frac{u_j - m_j}{u_j - l_j} \delta_{l_j} + \frac{m_j - l_j}{u_j - l_j} \delta_{u_j} \right]. \tag{2.1}$$

A singleton interval contributes the same atom to both measures. Zero weights are discarded. These are probability measures with the exact same mean as μ. They are physical mass-weighted bins, not equally weighted collocation points.

Theorem 2.1 (Two-sided edge-aware bracket)**.** *For every $x \geq 0$,*

$$R_{\mu^-}(x) \leq R_\mu(x) \leq R_{\mu^+}(x). \tag{2.2}$$

Consequently the same ordering holds for $G_{t,\cdot}(p)$ for every $p \in \mathcal{P}$, $0 \leq t \leq 1$, and for the variational values $\mathcal{F}_{\mu^-} \leq \mathcal{F}_\mu \leq \mathcal{F}_{\mu^+}$. For $\Delta = \sup_{0 \leq x \leq 1}(R_{\mu^+}(x) - R_{\mu^-}(x))$,

$$0 \leq \mathcal{F}_{\mu^+} - \mathcal{F}_{\mu^-} \leq \Delta/2. \tag{2.3}$$

If the partition is refined using the same actual measure, the lower brackets increase and the upper brackets decrease in convex order and in these transform and value orders. No resolvent gap is assumed here.

Proof. For every convex function f on the spectral interval, Jensen and the endpoint chord give $\int f \, d\mu^- \leq \int f \, d\mu \leq \int f \, d\mu^+$. Their upper support edges satisfy $e_- \leq e_\mu \leq e_+$: a bin with positive mass above a proposed edge contributes a positive upper endpoint weight. For $z > e_+$ the function $\lambda \mapsto (z - \lambda)^{-1}$ is convex, so $H_- \leq H_\mu \leq H_+$. More generally, if two such transforms obey this

order above the larger edge, their capped inverses have the same order. Indeed $b_{\mathrm{low}}(x) > b_{\mathrm{high}}(x)$ would put the former above both edges. It solves $H_{\mathrm{low}}(b) = x$, while strict decrease and the capped inverse convention give $H_{\mathrm{high}}(b) < x$, a contradiction. At zero the means agree. This proves (2.2). The uniform bound on the spectral terms survives adding the same S and taking the infimum, proving (2.3) without exchanging any optimizations. Within an old bin, Jensen applied to its sub-bin means raises the barycentric approximation. The old convex chord majorizes each sub-bin chord, so its endpoint approximation decreases. Apply the same argument to obtain the refinement statements. □

These are bounds on the explicitly defined functional independently of its identification with thermodynamic pressure. Their pressure interpretation requires the source's all-temperature equality. Monotonicity of a spectral resolvent is not by itself a proof of that equality.

3 Exact remainder and quadratic error control

For a bin let $v_j = \rho_j^{-1} \int (\lambda - m_j)^2 \mu_j(\, d\lambda)$. Write

$$V = \sum_j \rho_j (m_j - l_j)(u_j - m_j), \quad V_0 = \sum_j \rho_j v_j, \quad V_1 = V - V_0 \geq 0. \tag{3.1}$$

The inequality follows by averaging $(\lambda - l_j)(u_j - \lambda) \geq 0$. If the largest bin width is δ, then $V \leq \delta^2/4$.

Lemma 3.1 (Signed resolvent remainders)**.** *For $z > e_+$, the exact identities are*

$$H_\mu(z) - H_-(z) = \sum_j \int \frac{(\lambda - m_j)^2}{(z - \lambda)(z - m_j)^2} \mu_j(\, d\lambda), \tag{3.2}$$

$$H_+(z) - H_\mu(z) = \sum_j \int \frac{(\lambda - l_j)(u_j - \lambda)}{(z - l_j)(z - u_j)(z - \lambda)} \mu_j(\, d\lambda). \tag{3.3}$$

Singletons contribute zero. In particular, if every displayed denominator factor is at least $g > 0$, these errors are at most V_0/g^3 and V_1/g^3.

Proof. Expand the rational identity

$$\frac{1}{z - \lambda} = \frac{1}{z - m} + \frac{\lambda - m}{(z - m)^2} + \frac{(\lambda - m)^2}{(z - \lambda)(z - m)^2}.$$

The linear term integrates to zero in each bin. The difference between the endpoint chord of $(z - \lambda)^{-1}$ and its actual value is exactly the integrand of (3.3); multiplication by the three denominators verifies it. Its numerator integrates to $\rho_j[(m_j - l_j)(u_j - m_j) - v_j]$. This proves both claims. □

This is a measure-aware remainder in the sense of the freshly read original local-moment chapter [3]: a canceled first moment leaves a controlled second-order term, not an arbitrary polynomial exactness claim. Here rational algebra gives the full remainder without an infinite Taylor expansion.

Theorem 3.2 (A certified gap makes the bin error quadratic)**.** *Assume*

$$\gamma := b_-(1) - e_+ > 0. \tag{3.4}$$

Then $0 < \gamma \leq 1$ and, for $0 \leq x \leq 1$,

$$0 \leq R_\mu(x) - R_-(x) \leq \frac{xV_0}{\gamma^3}, \qquad 0 \leq R_+(x) - R_\mu(x) \leq \frac{xV_1}{\gamma^3}. \tag{3.5}$$

For every $p \in \mathcal{P}$, $t \in [0,1]$, the corresponding errors in G_t are bounded respectively by

$$\frac{t^2 V_i}{4\gamma^3}\left(1 - \int_0^1 p(s)^2\, ds\right), \qquad i = 0, 1. \tag{3.6}$$

In particular

$$0 \le \mathcal{F}_\mu - \mathcal{F}_{\mu^-} \le \frac{V_0}{4\gamma^3}, \quad 0 \le \mathcal{F}_{\mu^+} - \mathcal{F}_\mu \le \frac{V_1}{4\gamma^3}, \quad \mathcal{F}_{\mu^+} - \mathcal{F}_{\mu^-} \le \frac{\delta^2}{16\gamma^3}. \tag{3.7}$$

The same uniform errors propagate through the source's magnetic inner infimum and outer magnetization supremum, since only G depends on the spectral law. They also survive its external-field W_1 completion, conditional on that completion and its pressure identification.

Proof. The finite atomic lower law has an uncapped inverse. Write $b = b_-(x)$, $M_2 = \int (b - \lambda)^{-2} \mu^-(\, d\lambda)$, so $b' = -1/M_2$. For $c(x) = x(b_-(x) - e_+)$,

$$c'(x) = b - e_+ - x/M_2 \le 0, \qquad x - (b - e_+)M_2 = \int \frac{e_+ - \lambda}{(b-\lambda)^2} \mu^-(\, d\lambda) \ge 0.$$

Also $c(0+) = 1$. Thus $b_-(x) - e_+ \ge \gamma/x$, and all three inverses are above e_+ for $0 < x \le 1$. At $z = b_-(x)$, Lemma 3.1 gives $0 \le H_\mu(z) - x \le x^3 V_0/\gamma^3$. Until z reaches $b_\mu(x)$, Jensen implies $-H'_\mu(z) = \int (z - \lambda)^{-2}\, d\mu \ge H_\mu(z)^2 \ge x^2$. Integrating this derivative bounds the distance between the roots by xV_0/γ^3. Starting at $b_\mu(x) \ge b_-(x)$ and using the second remainder gives the other bound. Values at zero agree by the common mean.

The exact identity

$$\int_0^1 D_p(r)\, dr = \frac{1}{2}\left(1 - \int_0^1 p(s)^2\, ds\right) \tag{3.8}$$

follows by integrating $(p - r)_+$. Substitute $x = tD_p(r)$ into (3.5) and then integrate, proving (3.6). Uniform bounds survive adding any common field functional and taking infima, then any common outer supremum. The same argument and the bound $V \le \delta^2/4$ prove the remaining assertions. □

3.1 How to certify the premise

One need not subtract two poorly known large inverses. Choose a rational $z_0 > e_+$ with a rigorously evaluated $H_-(z_0) > 1$. Strict decrease then gives $b_-(1) > z_0$ and one may use any $0 < \gamma_0 \le z_0 - e_+$ in the estimates. Finite positive rational bins make this a rational inequality. Root bisection is monotone; verified intervals must also account for any uncertainty in the bin masses and means.

If the entire support interval has width $L < 1$, Jensen gives $b_-(x) \ge \int \lambda\, d\mu + 1/x$, hence the sufficient uniform gap $\gamma \ge 1 - L$. This is not necessary. More usefully at arbitrary fixed temperature, retain an actual atom of mass $a > 0$ at the upper edge as a singleton bin. Then $H_-(b) = x \ge a/(b - e_+)$, so $x(b_-(x) - e_+) \ge a$ and $\gamma \ge a$. This certificate survives any fixed scaling of the eigenvalues; the variance budget scales quadratically. No uniform zero-temperature error is inferred.

If (3.4) fails, Theorem 2.1 is still valid. One can compute its exact transform width using capped roots or refine bins. It is not legitimate to continue using a negative or unverified gap in (3.7). The physical-metric original [2] stresses the same distinction between a residual and a residual multiplied by a valid stability constant; no change of coordinates supplies that constant.

3.2 An explicit spectral example

For the uniform law on $[-a,a]$, $a>0$, one bin gives $\mu^- = \delta_0$ and $\mu^+ = (\delta_{-a}+\delta_a)/2$. Direct inversion gives

$$R_\mu(x) = a\coth(ax) - 1/x, \qquad R_+(x) = \frac{2a^2x}{1+\sqrt{1+4a^2x^2}}, \qquad R_- = 0, \tag{3.9}$$

with continuous values zero at $x=0$. Here $V=a^2$, $V_0=a^2/3$, $V_1=2a^2/3$. When $a<1$, $\gamma = 1-a$. With $a=1/4$, (3.7) yields $0 \le \mathcal{F}_\mu \le 1/81$ by comparison with the lower law. This is a certified bound on the variational value, not a formula for it.

For any $a>0$ an alternative explicit upper trial follows from $p=0$, $S(0)=0$, $D_0(r)=1-r$:

$$\mathcal{F}_\mu \le \mathcal{F}_{\mu^+} \le G_{1,\mu^+}(0) = \frac{1}{4}\left[\sqrt{1+4a^2} - 1 - \log\frac{1+\sqrt{1+4a^2}}{2}\right]. \tag{3.10}$$

Substitution $y = \sqrt{1+4a^2x^2}$ in $\frac{1}{2}\int_0^1 R_+(x)\,dx$ proves the formula. It is an upper candidate only, not an all-temperature replica-symmetric equality. Point spectra have $R_{\delta_a} = a$ and the exact finite-size pressure $a/2$ in zero field, consistent with all factors.

3.3 Why moments and bounded support still need edge tracking

Even matching mean and variance with the same support edges need not match the transform: the laws $\frac{1}{4}(\delta_{-2}+\delta_2)+\frac{1}{2}\delta_0$ and $\frac{1}{6}(\delta_{-2}+\delta_2)+\frac{1}{3}(\delta_{-1}+\delta_1)$ have mean zero and variance two, but at $z=3$ their resolvents are $7/15$ and $9/20$. These are exact finite facts, not a pressure counterexample.

A separate classical rank-one obstruction is stronger for the source hypotheses. Take $\Lambda_N = \mathrm{diag}(A,0,\dots,0)$, $A>0$, and let u be its Haar unit eigenvector in spin coordinates. All fixed empirical moments converge to those of δ_0, and all spectra lie in the same compact interval. Choosing $\sigma_i = \mathrm{sign}(u_i)$ in the partition sum gives

$$\mathbb{E}P_N \ge \frac{A}{2N}\mathbb{E}\|u\|_1^2 - \log 2 = \frac{A}{2N} + \frac{A(N-1)}{\pi N} - \log 2. \tag{3.11}$$

Indeed Gaussian radial independence gives $\mathbb{E}|u_1u_2| = \mathbb{E}|G_1G_2|/\mathbb{E}\|G\|^2 = 2/(\pi N)$. For $A > \pi\log 2$, the limit inferior in (3.11) is positive, whereas $\mathcal{F}_{\delta_0} = 0$. The missing premise is exactly upper-edge convergence. This bounded-spike illustration supplements the source's unbounded negative-spectrum counterexample; it has no historical-novelty claim.

4 A sharper constrained-block replacement

This second result concerns the actual magnetic proof in source `70-magnetic.tex`, particularly its Lemma `mag:replacement`. It is not a new proof of the thermodynamic cavity comparison. Fix positive rational group proportions γ_j, $1 \le j \le J$, summing to one, interior rational magnetizations $m_j \in (-1,1)$, and a height cap $0 \le H < \infty$. For each admissible n, let $n_j = n\gamma_j$ and $k_j = n_j(1+m_j)/2$ be positive-group integer counts. Let $\mathcal{C}_n$ be the spin block with exactly k_j plus spins in group j. The constraint keeps its original mass $2^{-n}\mathbf{1}_{\mathcal{C}_n}$; normalizing that mass away would change the functional.

For $h \in \mathcal{H}$ with $h(1-) \le H$, let $\ell_b^{\mathrm{can}}(h) = F_{-1}(b) - h(1-)/2$ and define

$$\ell^{[\mathbf{m}]}(h) = \sum_j \gamma_j \inf_{b\in\mathbb{R}}\{\ell_b^{\mathrm{can}}(h) - bm_j\}. \tag{4.1}$$

Let $\ell^{(n)}(h)$ be the n-coordinate version of the Gaussian recursion with terminal payoff

$$\log\left(2^{-n}\sum_{\varepsilon\in\mathcal{C}_n} e^{z\cdot\varepsilon}\right),$$

subtracting $nh(1-)/2$ and dividing by n. Write $B_i^*(h)$ and $B_i^{(n)}(h)$ for the group-weighted canonical and site-averaged constrained squared conditional spin means at level i. Equivalently they are the expected same-site product of two spin draws whose Gaussian paths share that level. Even two visits to the same leaf have separate terminal spin draws. These are the source's quantities, with the same fixed cascade weights in both recursions.

4.1 A depth-free tail estimate with explicit constants

Set

$$\begin{aligned} m_* &= \max_j |m_j| < 1, \quad B = \frac{H/2+\log 2}{1-m_*}, \quad A = B+H, \quad V_H = \max(1,H), \\ d_n &= \frac{1}{2}\sum_j \sqrt{n_j(1-m_j^2)} \le \frac{1}{2}\sqrt{Jn} \le n/2. \end{aligned} \tag{4.2}$$

The harmless enlargement $V_H \ge 1$ includes the zero-height case. Let $\mathsf{h}(v) = -v\log v-(1-v)\log(1-v)$, with endpoint values zero.

Theorem 4.1 (Quantitative block replacement)**.** *For every admissible n, every $u>0$, and every finite nondecreasing path of height at most H, define*

$$\Lambda_n(u) = 2(A+u)d_n + 4n(A+u+V_H/u)e^{-u^2/(2V_H)} + n\mathsf{h}(d_n/n). \tag{4.3}$$

Then, uniformly in the number and weights of the cascade levels,

$$0 \le L_n(h) := n\big(\ell^{[\mathbf{m}]}(h) - \ell^{(n)}(h)\big) \le \Lambda_n(u), \tag{4.4}$$

$$\max_i |B_i^{(n)}(h) - B_i^*(h)| \le \min\{1,\, 2\sqrt{L_n(h)/n}\} \le \min\{1,\, 2\sqrt{\Lambda_n(u)/n}\}. \tag{4.5}$$

With $u=\sqrt{V_H\log(n+1)}$, this gives

$$\sup_h |\ell^{(n)}-\ell^{[\mathbf{m}]}| = O_{H,\mathbf{m},J}(n^{-1/2}\log(n+1)), \qquad \sup_{h,i} |B_i^{(n)}-B_i^*| = O_{H,\mathbf{m},J}(n^{-1/4}\sqrt{\log(n+1)}). \tag{4.6}$$

The constants do not require a positive lower bound on the step weights. No uniformity as $m_*\uparrow 1$ or $H\to\infty$ is asserted.

Proof. We first give the canonical measure and a tail bound, then compare its constraint loss and two-path law. Let b_j be the unique canonical bias prescribing mean m_j. Existence and uniqueness follow either from the source's bias argument or directly from strict convexity of the backward value $f_h(b) = F_{-1}(b)$ and $f_h'(b)\to\pm 1$. For completeness, convexity and comparison with exponent one give $\log\cosh b \le f_h(b) \le \log\cosh b + H/2$. At its minimizer, $f_h(b_j) - b_j m_j \le f_h(0) \le H/2$. Since $f_h(b_j) \ge |b_j| - \log 2$, it follows that $|b_j| \le B$.

Insert these biases in the canonical and constrained n-coordinate recursions. Every canonical backward value adds over sites. A canonical increment g of variance v, conditional on its past

position z, has density proportional to $e^{\zeta F(z+g)}\gamma_v(\,dg)$, with $0 \le \zeta \le 1$ and F 1-Lipschitz. Gaussian shifting proves

$$\mathbb{E}_{\mathrm{can}}(e^{tg} \mid \text{past}) = e^{t^2 v/2}\frac{\mathbb{E}e^{\zeta F(z+g+tv)}}{\mathbb{E}e^{\zeta F(z+g)}} \le e^{t^2 v/2+|t|v}. \tag{4.7}$$

The root is an ordinary Gaussian and satisfies the sharper inequality without $|t|v$. Successive conditioning, summing variances, and including the initial bias yield for each terminal field Z_r

$$\mathbb{E}_{\mathrm{can}}e^{tZ_r} \le e^{|t|A+t^2V_H/2}, \qquad \Pr_{\mathrm{can}}(|Z_r| > A+v) \le 2e^{-v^2/(2V_H)} \quad (v>0). \tag{4.8}$$

No union bound over levels occurs. Integrating the last tail gives

$$\mathbb{E}_{\mathrm{can}}[|Z_r|\mathbf{1}_{\{|Z_r|>A+u\}}] \le 2(A+u+V_H/u)e^{-u^2/(2V_H)}. \tag{4.9}$$

Here $\int_u^\infty e^{-v^2/(2V_H)}\,dv \le (V_H/u)e^{-u^2/(2V_H)}$ follows by bounding $v/u \ge 1$.

Under the averaged canonical path and final-spin law, sites are independent and the mean spin in group j is m_j; this follows from the factorized recursion and its bias derivative. The plus count in that group has binomial variance $n_j(1-m_j^2)/4$. A deterministic repair to the required counts flips exactly $D = \sum_j |K_j - k_j|$ spins, so $\mathbb{E}_{\mathrm{can}}D \le d_n$. The repair may depend on the spin configuration and fields. Independence of its chosen flip set and the fields is neither assumed nor needed: pointwise, with S its flip set,

$$\sum_{r\in S}|Z_r| \le (A+u)D + \sum_{r=1}^{n}|Z_r|\mathbf{1}_{\{|Z_r|>A+u\}}.$$

Flipping a spin changes its linear exponent by at most $2|Z_r|$. Thus its expected absolute exponent cost is at most the first two terms of (4.3).

At fixed fields, push the canonical Gibbs spin law through this repair. The elementary Gibbs variational identity bounds the constrained terminal log-partition loss by its expected exponent cost plus the entropy lost by the map. The original spin is determined by its image and flip mask. Conditional entropy loss is therefore at most mask entropy. Entropy subadditivity and concavity, followed by averaging, bound this by $n\mathsf{h}(\mathbb{E}D/n) \le n\mathsf{h}(d_n/n)$; the last inequality uses $d_n/n \le 1/2$. The factor 2^{-n} is the same in both partitions and cancels in this loss, rather than being renormalized on the constraint.

Let U_i, V_i be the canonical and constrained biased backward values and $\Delta_i = U_i - V_i \ge 0$. The root loss is $\mathbb{E}\Delta_0 = L_n$; biases are constant on the constraint, which explains the equality with (4.4). At a positive-exponent step the canonical kernel gives

$$\Delta_{i-1} = -\zeta_i^{-1}\log \mathbb{E}_{\mathrm{can}}e^{-\zeta_i\Delta_i} \le \mathbb{E}_{\mathrm{can}}\Delta_i.$$

Iteration transfers the terminal loss bound to the root and proves $0 \le L_n \le \Lambda_n(u)$.

For clarity, we also give the entropy step, rather than replacing the constraint by an unconditional average. The ratio of constrained to canonical mark kernels at the *same past* is $e^{\zeta_i(\Delta_{i-1}-\Delta_i)}$. At the terminal spin draw it is $e^{\Delta_{k-1}}$ on allowed states and zero elsewhere. Root mark laws coincide. Therefore conditional entropy, with its outer average under the constrained past, telescopes as

$$\sum_{i>0}\zeta_i^{-1}\mathbb{E}_{\mathrm{con}}\,\mathrm{KL}(K_i^{\mathrm{con}}\|K_i^{\mathrm{can}}) + \mathbb{E}_{\mathrm{con}}\,\mathrm{KL}(K_{\mathrm{spin}}^{\mathrm{con}}\|K_{\mathrm{spin}}^{\mathrm{can}}) = L_n. \tag{4.10}$$

Since $\zeta_i \le 1$, ordinary one-path entropy is at most L_n. Given a specified common level of two sampled paths, shared kernels are counted once and post-split kernels at most twice. The marked-cascade law preserves that branching law and the one-path marginals, as in source Proposition

`arr:rpc`. Thus the two-path entropy is at most $2L_n$. This finite cascade marking premise is exactly the source mechanism; no independence of the constrained sites or of their common histories is used.

Under the canonical two-path law conditional on the branching level, sites are independent. If $X_r = \varepsilon_r^1 \varepsilon_r^2 \in [-1, 1]$, then $n^{-1} \sum_r \mathbb{E}_{\mathrm{can}} X_r = B_i^*$ and Hoeffding's lemma gives

$$\log \mathbb{E}_{\mathrm{can}} e^{t \sum_r (X_r - \mathbb{E}_{\mathrm{can}} X_r)} \le nt^2/2.$$

For probability laws Q, P, Jensen applied after changing measure proves $\mathbb{E}_Q Y \le \mathrm{KL}(Q\|P) + \log \mathbb{E}_P e^Y$. Apply it with both signs of t, divide by nt, and minimize over $t > 0$. With $\mathrm{KL}(Q\|P) \le 2L_n$, this yields $|B_i^{(n)} - B_i^*| \le 2\sqrt{L_n/n}$. Both quantities belong to $[0, 1]$, giving the minimum with one. The case $L_n = 0$ follows by letting $t \downarrow 0$. Finally choose the stated u, so the exponential equals $(n+1)^{-1/2}$, and use $\mathsf{h}(v) \le v \log(e/v)$ for $v \le 1/2$. All constants depend only on the displayed fixed data, proving the rates. □

Corollary 4.2 (Explicit supporting-plane replacement)**.** *Let a test field v have height at most $A_0 < \infty$ and h height at most H. Retain the source's concavity/supporting inequality for $\ell^{[\mathbf{m}]}$. Then*

$$\ell^{[\mathbf{m}]}(v) \le \ell^{(n)}(h) - \frac{1}{2} \int B_h^{(n)}(v - h) + \frac{\Lambda_n(u)}{n} + (A_0 + H)\sqrt{\frac{\Lambda_n(u)}{n}}. \tag{4.11}$$

The error is uniform in the common step refinement, but not in A_0, H.

Proof. Replace $\ell^{[\mathbf{m}]}(h)$ and B_h^* in the source supporting inequality by their block counterparts using (4.4) and (4.5). The integral of $|v - h|$ is at most $A_0 + H$. □

This improves the displayed estimate in source `mag:loss-rate`, which uses Cauchy–Schwarz with $\mathbb{E}\|Z\|^2 = O(n)$ and $\mathbb{E}D = O(\sqrt{n})$ to obtain $n^{3/4}$. The improvement is a tail-sensitive bound for the *selected* flip coordinates. The source already has the needed Gaussian moment-generating bound; it is not a newly discovered concentration theorem. Likewise the weighted entropy identity is retained from its proof. The new quantitative two-spin bound uses it directly with exponential moments instead of only showing vanishing deviation probabilities.

4.2 An exact benchmark and the order of limits

At $h = 0$, one group and $m = 0$, even n gives exactly

$$L_n = \log \frac{2^n}{\binom{n}{n/2}}. \tag{4.12}$$

The canonical bias is zero, all fields vanish and only the slice mass remains. This example has much smaller logarithmic loss; the uniform bound above is not claimed optimal. It also detects an incorrect convention that would normalize the constrained prior and falsely return zero loss.

In the source lower bound, the system size N tends to infinity at a fixed admissible block size n, then n grows, and the test height cap grows later. Theorem 4.1 quantifies only the middle replacement. The spectral finite-alphabet field height can depend on that alphabet. Nothing here allows all caps, spectral bins, temperature and N to vary simultaneously without further uniform estimates.

5 Evaluation, comparisons and remaining obligations

5.1 A cancellation-free root near zero

The preceding formulas are exact; numerical subtraction of $1/x$ from $b_\mu(x)$ can nevertheless be ill conditioned as $x \downarrow 0$. For support in $[l, u]$, write $L = u - l$ and suppose $xL < 1$, $x \geq 0$. The same transform is the unique root $R \in [l, u]$ of

$$F_x(R) = \int \frac{\lambda - R}{1 + x(R - \lambda)} \mu(\, d\lambda) = 0, \qquad \partial_R F_x(R) = -\int \frac{\mu(\, d\lambda)}{[1 + x(R - \lambda)]^2}. \tag{5.1}$$

Indeed $F_x(l) \geq 0$, $F_x(u) \leq 0$, and the derivative is strictly negative. For $x > 0$, $xF_x(R) = H_\mu(R + 1/x)/x - 1$; the resulting inverse is strictly above u because $R + 1/x \geq l + 1/x > u$. At $x = 0$ the root is the mean. For any $\widetilde{R} \in [l, u]$,

$$|\widetilde{R} - R_\mu(x)| \leq (1 + xL)^2 |F_x(\widetilde{R})|.$$

If moments are used to evaluate the root, the finite polynomial $\sum_{k=0}^{m} x^k \int (\lambda - \widetilde{R})^{k+1} \mu(\, d\lambda)$ has residual error at most $L(xL)^{m+1}/(1 - xL)$, by the exact geometric remainder. This also accommodates atomic measures. The point-law case $L = 0$ is exact. These are elementary evaluation identities, not an additional novelty claim or an assertion of measured numerical speedup. They do not replace the capped definition outside their stated domain.

5.2 What the refinements add

The source spectral approximation proves qualitative uniform continuity, uses a $1/2$ operator-norm pressure comparison for matched matrices, and retains spectral extremes. Our bins instead produce an ordered pair of variational values. For a verified gap, preserving each bin's first moment improves the scalar transform error to a variance-weighted quadratic bound. No spectral density is assumed. The gain has a cost: actual bin masses and means, an upper support bound, and a verified inverse gap must be available. Uncertified empirical moments are not substitutes. The gap-free order theorem is weaker quantitatively but remains usable when the stability constant deteriorates.

The block improvement is different: it uses the actual conditional Gaussian mark law to control the energy of a selected repair set, then propagates weighted relative entropy to a two-spin observable. Its constants are uniform in finite cascade depth at fixed height, not in height, temperature, near-boundary magnetizations or the original system size. It avoids assuming independence under the constrained law. This distinction matters because conditioning on exact group counts couples the coordinates.

The freshly read private physical-metric chapter [2] supplies an original explanation of measure weights, omitted components and stability constants. Its Hilbert Gram identities are not applied as positivity certificates for the nonlinear quenched pressure. The original local-moment chapter [3] supplies the centered remainder and truncation viewpoint; its included subsidiary chapters were not read here. Our displayed rational remainder identities were proved directly. Neither private chapter contains a proof of the Haar cavity equality.

5.3 Current public context and future work

Primary version records checked on October 9, 2026 distinguish several scopes. Fan–Wu [4] gives a high-temperature replica-symmetric result with external field. Fan–Misiakiewicz–Wang–Wen [5] gives dynamical mean-field and replica-symmetric results under a rapid-mixing analysis, with the

explicit Ising operator-norm condition $\|J\|_{\mathrm{op}} < 1/2$ in its abstract. Neither abstract asserts the source's all-temperature formula. Bhattacharya–Sen [6] concerns high-temperature orthogonal models; its displayed Hamiltonian convention does not include our factor 1/2. Guionnet–Maida [7] concerns finite-rank spherical integrals and the R transform, not by itself the full quenched Ising variational identity. Ho's concavity preprint [8] is relevant context for the field functional. These external proofs were not independently audited; the fresh checks cover abstracts and version metadata only.

Useful next steps are adaptive bins minimizing the certified variance budgets, interval arithmetic for edge-root brackets, and a sharper repair argument exploiting more than one-coordinate tails. Extending the block rate uniformly as the height cap grows would require new tail and bias estimates. A simultaneous thermodynamic and discretization limit requires uniform control of the source Haar and cavity errors. No such uniformity follows from this report's finite bounds.

6 Verification and assessment

The accompanying exact checker uses rational arithmetic for positive atomic resolvent identities, bin variances, normalized-gap certificates, the overlap integral, moment counterexamples and finite constrained-spin laws. It also checks selected-coordinate bounds without independence. Its receipts distinguish parameter cases from assertions; these fixtures are supplements to the proofs, not tests of the infinite Haar theorem. The pre-read and post-read ledgers record all literal intervals and file hashes. Parent checks are separate and are not included in the author's test totals.

The final owned run passed 22,050 cases and 38,411 exact assertions:

Fixture family	Cases	Assertions
Atomic bin remainders	972	7,776
Certified inverse brackets	108	729
Overlap integral	251	502
Stable-root finite remainders	1,215	3,645
Exact count repairs	3,076	9,228
Binomial repair moments	28	112
Selected-coordinate bound	16,384	16,384
Worked moment and variance fixtures	1	5
Zero-height slice mass	15	30

Root fixtures use rational bisection intervals, not floating-point approximations asserted to be exact. The checker is independently executable from `_audit/M447-checks.py`.

Applicability remains L3. The earned assessment is C3: the ordered spectral enclosure with a checkable quadratic error and the fixed-cap constrained-block rate are notable local quantitative increments. Convex order, geometric remainders, Gaussian exponential moments, relative entropy and the rank-one obstruction are classical tools, not claimed discoveries. The source global formula remains attributed, not independently certified.

References

[1] OpenAI mathematics repository, *All-temperature pressure for orthogonally invariant Ising spin glasses*, September 25, 2026. Pinned public source subtree. All ten cached text files read; byte identity is provenance, not proof validation.

[2] M. Subbarao, original private chapter *Physical metrics*, RamCalBooks, volume 21, chapter file `ch01d-physical-metrics.tex`, lines 1–210. Fresh local literal read; no private material was uploaded.

[3] M. Subbarao, original private local-moment chapter, RamCalBooks, volume 1-rev2, `ch03-local-moments-rev2.tex`, lines 1–446. Fresh local literal read of this file, not its subsidiary inputs.

[4] Z. Fan and Y. Wu, *The replica-symmetric free energy for Ising spin glasses with orthogonally invariant couplings*, arXiv:2105.02797, v3, April 30, 2024. Abstract and version record checked.

[5] Z. Fan, T. Misiakiewicz, L. Wang and G. G. Wen, *Dynamical mean-field limit and replica-symmetric free energy for the orthogonally-invariant SK model*, arXiv:2607.10102, v2, September 21, 2026. Abstract and version record checked.

[6] B. B. Bhattacharya and S. Sen, *High temperature asymptotics of orthogonal mean-field spin glasses*, arXiv:1507.05067, v2, August 20, 2015. Abstract and version record checked.

[7] A. Guionnet and M. Maida, *An asymptotic log-Fourier interpretation of the R-transform* (arXiv title), arXiv:math/0406121, v3, March 11, 2005. Abstract and version record checked.

[8] F.-H. Ho, *A Concavity Theorem for the Parisi PDE*, arXiv:2606.15432, v1, June 13, 2026. Abstract and version record checked.